

Multiple Choice

1. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

2. **Answer:** A

Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9

Note: Correct option: A - decreased by a factor of 9

3. **Answer:** A

Options: A) Planck's constant; B) Boltzmann's constant; C) The Rydberg constant; D) The speed of light

Note: Correct option: A - Planck's constant

4. **Answer:** A

Options: A) a dye laser; B) a helium-neon laser; C) an excimer laser; D) a ruby laser

Note: Correct option: A - a dye laser

5. **Answer:** C

Options: A) $1/2$; B) 1; C) $3/2$; D) $5/2$

Note: Correct option: C - $3/2$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '250'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hall coefficient'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'

Long Form Response

11. **Answer:** $2 \times 10^{-5} \text{ N}$. *Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.*

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

11. **Answer:** 0.5 mm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key